

Code-Layout, Readability and Reusability

Date	15 July 2026
Team ID	SWTID-2026-7750
Project Name	Credit card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation used throughout the code.	Yes	Code follows Python indentation standards.
2	Proper File Structure	Files and folders are logically organized.	Yes	Project is organized into app.py, prepare_data.py, train_model.py, templates, static, and dataset folders.
3	Meaningful Variable Names	Variables clearly describe their purpose.	Yes	Descriptive names such as df, model, feature_names, encoders, probability, and prediction are used.
4	Function / Method Names	Functions are named based on their functionality.	Yes	Functions are easy to understand and follow Python naming conventions.
5	Code Comments	Inline comments explain important sections of the code.	Partial	Comments are included for major steps, but additional comments can improve readability.
6	Modular Design	Code is divided into reusable modules and files.	Yes	Data preparation, model training, and web application are implemented in separate Python files.
7	No Redundant Code	Duplicate or unnecessary code has been removed.	Yes	Code is organized efficiently with minimal duplication.
8	Error Handling	Exceptions and invalid inputs are handled properly.	Partial	Basic validation is available; additional exception handling can be added for robustness.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Preprocessing Module	Python, Pandas	prepare_data.py and train_model.py	High
2	Feature Engineering Module	Python, Pandas	Data preprocessing and prediction	High
3	Label Encoding Module	Scikit-learn	Training and prediction	High
4	Random Forest Model	Scikit-learn	Model training and Flask application	High
5	Flask Web Application	Python, Flask	User prediction interface	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project folders and files are well organized and easy to navigate.
Readability	4	Code is readable with meaningful names; additional comments could improve clarity.
Reusability	5	Modules such as preprocessing, feature engineering, and model loading can be reused in future projects.
Documentation / Comments	4	README documentation is provided, but more inline comments could be added.
Overall Score	4.5 / 5	The project follows good coding practices, modular design, and maintainable structure suitable for academic and future enhancements.